

Introduction to Computer Vision(CS708)

Lecture01 - Introduction

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Introduction

Vision systems



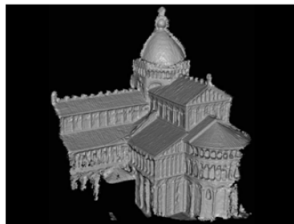
Figure: The human visual system has no problem interpreting the subtle variations in translucency and shading in this photograph and correctly segmenting the object from its background..

Introduction

Vision algorithms and application.



(a)



(b)

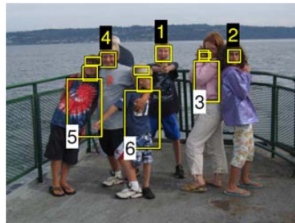
Figure: (a) **Structure from motion** algorithms can reconstruct a **3D point model** of a large complex scene from hundreds of partially overlapping photographs (b) **Stereo matching** algorithms can build a detailed 3D model of a building facade from hundreds of differently exposed photographs taken from the Internet.

Introduction

Vision algorithms and application.



(c)



(d)

Figure: (c) Person tracking algorithms can track a person walking. (d) In front of a cluttered background, Face detection algorithms, coupled with color-based clothing and hair detection algorithms, can locate and recognize the individuals in this image.

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2 Applications

3 About course

- Software used
- Marks distribution
- Readings

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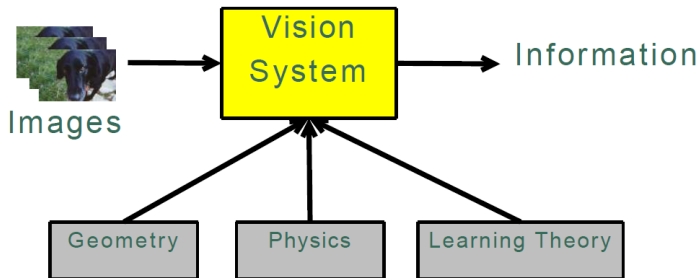


Figure: Source: Dr. Matthew

This kind of information we want is application specific:

- 3D models
- Object poses
- Object categories
- Camera poses

Introduction

Parts of system

The “vision system” includes

- Image **acquisition** hardware
 - Analog camera plus digital frame grabber, -or-
 - Digital camera with a fast serial interface (Firewire, USB, etc.)
- Image processing **support software**
- Your computer vision **algorithms**

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Application

Important application include

- Optical character recognition (OCR)
- Industrial inspection and control
- Mobile robot navigation
- Military intelligence
- Security (Surveillance)
- Fingerprint recognition and biometrics
- Image retrieval from digital libraries
- Medical image analysis
- 3D model capture for visualization and animation

Introduction

Broader consumer-level applications

Important broader consumer-level applications include

- Stitching
- 3D modeling
- Face detection
- Visual authentication
- Morphing
- Exposure bracketing
- Photo based walkthroughs

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Vision systems



(a)



(b)

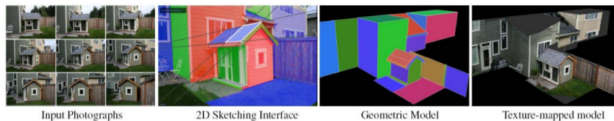
Figure: (a) image stitching: merging different views. (b) exposure bracketing: merging different exposures.

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Vision systems



(c)



(d)

Figure: (c) morphing: blending between two photographs. (d) turning a collection of photographs into a 3D model.

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The software used during the course

- Python 3.x
- OpenCV Libraries

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Marks distribution

- Mid-term (30%)
- Terminal (40%)
- Assignments and Quizzes (15%)
- Research Report (15%)

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Software used

Textbooks and resources used in this course

- Computer Vision: Algorithms and Applications by *Richard Szeliski*
- Introduction to Computer Vision - CS143
- Computer vision from Mubarak Shah

Introduction

This semester

This semester we focus on algorithms for

- 3D reconstruction
- Learning
- Sequential state estimation (e.g. tracking)

To understand modern 3D reconstruction techniques we need to understand how **cameras** transduce the world into image.

This involves understanding **projective geometry** and camera models.

Then we can begin to figure out how to invert what the camera does or **reconstruct the 3D scene**

The End